MACHINE LEARNING AND DEEP LEARNING · EXAM CASE

Who will leave in the next 30 days?

Churn prediction for a subscription business

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1 · THE PROBLEM

The goal: a ranked list of customers at risk

PREDICT

Will this customer cancel within 30 days?

Yes or no: binary classification

ONE ROW

One customer on one date

Snapshot 2026-08-01; only data known that day may be used

DECISION

Whom to contact first?

The retention team has limited capacity, so ranking matters most

2 · THE DATA

1,800 customers, and only 18% churn

18%

churn: always guessing “stays” scores 82% accuracy and finds no one

PROBLEMS FOUND AND FIXED

18 exact duplicates

Removed before the split, so no customer is in both train and test

79 impossible rows

0 login days but 18-minute sessions: session time set to missing

NPS with decimals

Probably an average, so interpreted with care

3 · LEAKAGE

A higher score can be a fake score

days_until_churn

Only filled in for customers who already left. It gives away the answer, so it was dropped.

retention_offer_sent

65% churn with an offer vs 8% without. Offers are a reaction to known churn, and they are the decision the model should support. Excluded.

ROC-AUC, LOGISTIC REGRESSION

0.86 with the offer: fake

0.75 without: honest

I chose the lower, honest score.

4 · METHOD

A fair, leak-proof setup

1

Split 80 / 20

Stratified. The test set is locked away and used once, at the end.

2

Pipeline

Imputing, scaling and encoding learn from training data only.

3

5-fold CV

The same folds for every model, so the comparison is fair.

4

Two baselines

Always “stays”, and a business rule: monthly contract + few logins.

5 · MODEL COMPARISON

The simplest model wins; complex ones overfit

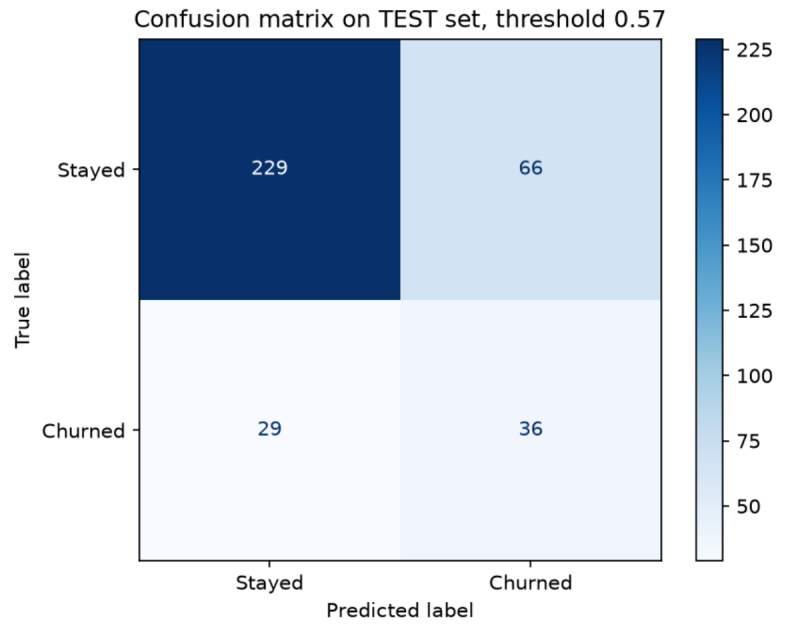
Model (5-fold CV)	ROC-AUC	PR-AUC	Train ROC-AUC	Overfit gap
Logistic Regression (best)	0.753	0.429	0.797	0.04 (small)
Random Forest	0.747	0.409	0.992	0.25 (large)
Gradient Boosting	0.720	0.372	0.952	0.23 (large)
Business rule (baseline)	0.618	0.255	–	–

Tuning (grid search) did not change the ranking: the data is small and the patterns are simple.

Age removed: ROC-AUC 0.753 → 0.752. Same performance, no age-based targeting.

6 · RESULTS AND ERRORS

Test ROC-AUC 0.76: 2.1× better than random



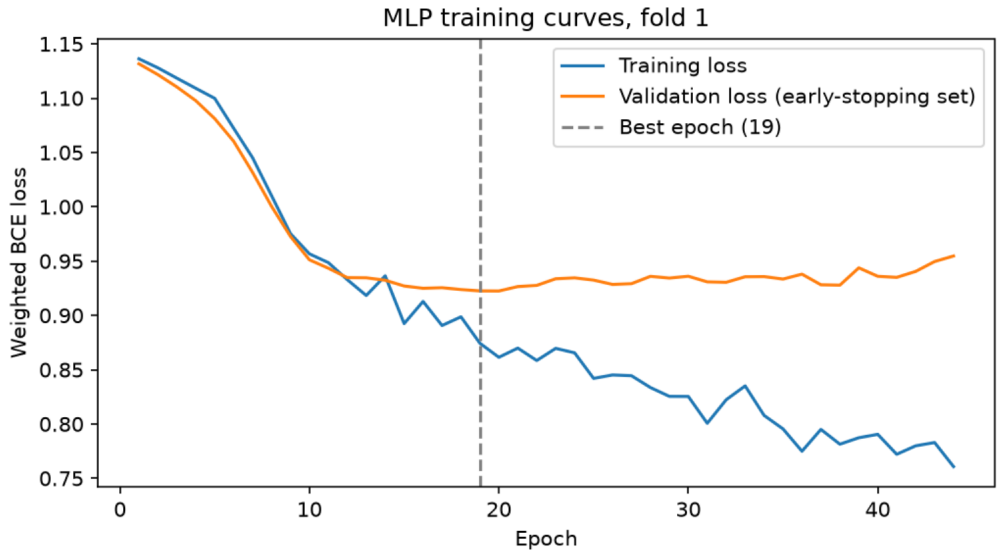
55% of churners found (36 of 65) at threshold 0.57

2.1× more churners in the top 20% than random

Missed churners look loyal

14 login days a month, vs 8 for churners found. The fix is new data, not a bigger model.

Deep Learning: tested, no gain



PyTorch network

31 → 32 → 16 → 1, ReLU, dropout, Adam, early stopping

Same as Logistic Regression

ROC-AUC 0.749 vs 0.752; won 2 of 5 folds. Overfits after ~15 epochs.

Transfer Learning: not relevant

Our columns are company-specific. It would fit support-ticket text with a pretrained language model.

8 · RECOMMENDATION

Logistic Regression as a monthly risk ranking

Why

- As good as or better than complex models
- Least overfitting, stable
- Shows why a customer is flagged

Limits

- One snapshot in time
- Small data: ~320 churners
- Weaker on long contracts

Next

- Test on a later snapshot
- Add months until renewal
- A/B test retention offers

IN PRODUCTION

Pipeline saved with input schema and versions · validate new data · monitor drift and recall (true labels arrive 30 days late) · retrain on drift or quarterly

Thank you. Questions?

An honest 0.75 that holds up in practice beats a 0.86 that doesn’t.